2

Data Collection

April, 2026

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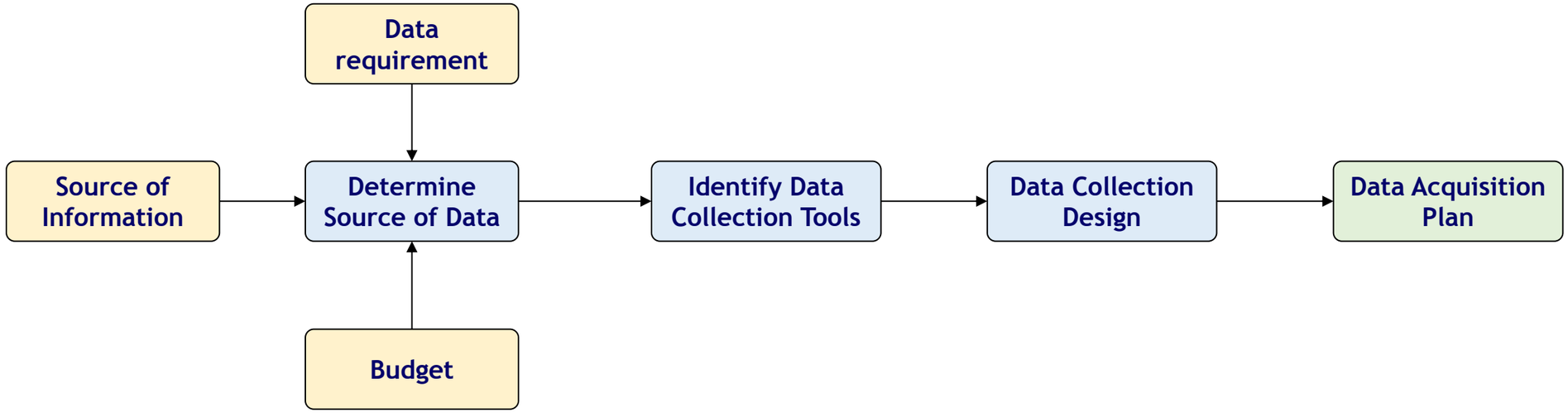
Definition

Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes.

Factors to be Considered Before Collection of Data

- Object and scope of the enquiry.
- Sources of information.
- Quantitative expression.
- Techniques of data collection.
- Unit of collection.



Internal & External Sources of Data

Internal sources of Data

- Information institutions and departments have about their regular functions for internal purposes.
- When this information is used in a survey, it's called internal data (e.g., social welfare societies).

External sources of data

- Information collected from outside agencies.
- This data is either primary or secondary and can be collected via census or sampling methods through surveys.

Primary Data

- Data collected from first-hand experiences.
- It is considered more reliable, authentic, and has not been published elsewhere.
- Its validity is greater than secondary data as it hasn't been altered. Includes methods like questionnaires.

Collection methods

- Observation: Investigation through observation.
- Interview: Direct Personal Investigation (i.e., interview method).
- Questionnaire: Indirect oral investigation (i.e., through enumerators).
- Case Study: Investigation through mailed questionnaire.
- Survey: Investigation through local reporters/questionnaire.

Secondary Data

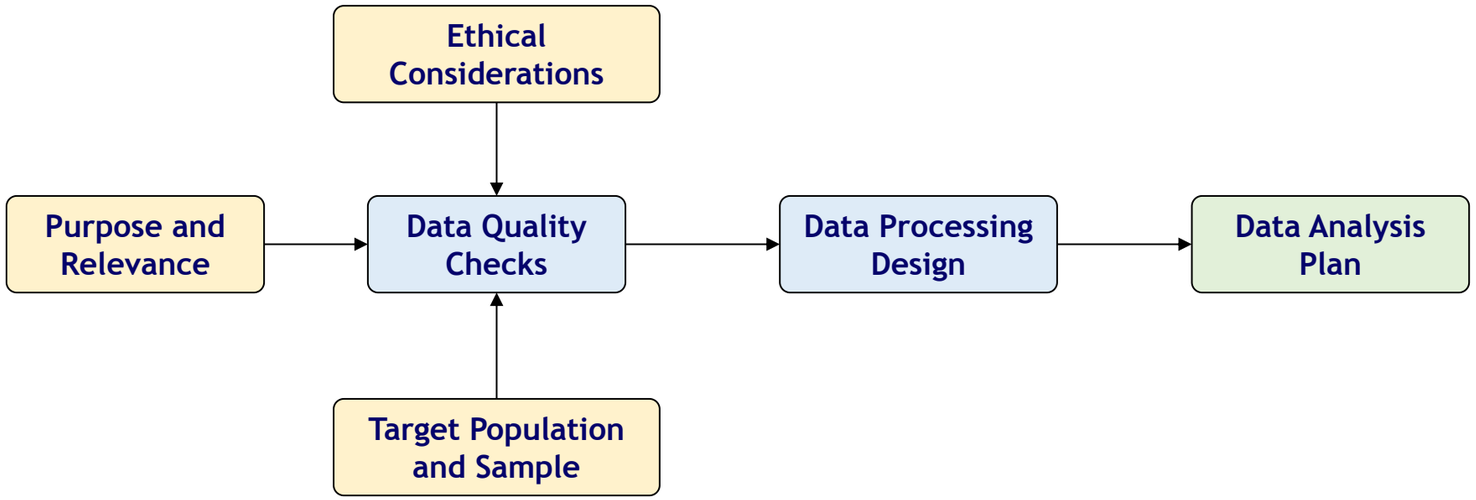
- Data that has already been collected by others.
- Usually found in journals, periodicals, research publications, official records, etc.
- Used when primary data collection is not possible.
- Collected for a purpose other than the current research problem.

Sources of data

- Public records, publications of state , local government
- Books, magazines, newspaper
- Reports & publications of industry ,bank, stock exchange
- Reports by research scholars, universities, economist

Factors to be Considered Before Using Data

- Purpose and Relevance
- Data Quality
- Ethical Considerations
- Target Population and Sample
- Timeliness and Resources



Tools & Libraries - API Tools

Purpose	Tool/Library	Description
Sending HTTP requests	`requests`	The most commonly used library to interact with REST APIs.
Advanced HTTP client	`httpx`	Async support, HTTP/2, and more features than requests.
API specification	`OpenAPI, Swagger, Postman`	Useful for documenting and testing APIs.
Handling GraphQL	`gql, sgqlc`	Tools to query GraphQL APIs using Python.
OAuth 2.0 support	`requests-oauthlib`	For interacting with APIs requiring OAuth (e.g., Google, LinkedIn).

Tools & Libraries - Web Scraping

Purpose	Tool/Library	Description
HTML parsing	`BeautifulSoup`	Easy-to-use HTML parser for scraping web data.
Automated browser	`Selenium`	Controls a browser (Chrome, Firefox) for scraping JS-heavy sites.
Fast async scraping	`aiohttp, httpx, scrapy`	Used for high-performance or parallel scraping.
Headless browser	`Playwright, pyppeteer`	Handle modern JS-rich websites.

Example: Use Open-Meteo API to get hourly temperature data

- We'll request hourly temperature forecast for Paris (latitude: 48.8566, longitude: 2.3522):

```python
import requests

# Define the endpoint and parameters
api_url = "https://api.open-meteo.com/v1/forecast"
params = {
    "latitude": 48.8566,
    "longitude": 2.3522,
    "hourly": "temperature_2m",
    "timezone": "auto"
}

# Send the GET request
response = requests.get(api_url, params=params)

# Parse and print
if response.status_code == 200:
    data = response.json()
    times = data['hourly']['time']
    temps = data['hourly']['temperature_2m']
    print("Next 5 hourly temperatures in Paris:")
    for t, temp in zip(times[:5], temps[:5]):
        print(f"{t}: {temp}°C")
else:
    print("Failed to fetch data:", response.status_code)
```

Example: Scrape documentation titles from the Open-Meteo website

- We'll scrape the Open-Meteo docs page to get the section headers https://open-meteo.com/en/docs :

```python
from bs4 import BeautifulSoup
import requests

url = "https://open-meteo.com/en/docs"
response = requests.get(url)

if response.status_code == 200:
    soup = BeautifulSoup(response.text, "html.parser")

    print("Section headers from the documentation page:")
    for header in soup.find_all(["h2", "h3"]):
        print("-", header.get_text(strip=True))
else:
    print("Failed to scrape the website:", response.status_code)
```

Practical Lab: Web Scraping Company Data from Societe.com

Objective

- Create a Python script to extract yearly turnover (CA) and number of employees from a company profile page on societe.com.

Ethical Note

- Societe.com is a commercial website.
- Make sure scraping respects robots.txt and use the data only for ***educational purposes***.
- Save pages (cache) locally to simulate extraction.

Tools

- `requests, beautifulsoup4`

References

https://www.scribbr.com/methodology/data-collection/
https://www.slideshare.net/slideshow/data-collection-presentation-34042800/34042800#3